FINC5001

Workshop 4

Debt and Equity Valuation

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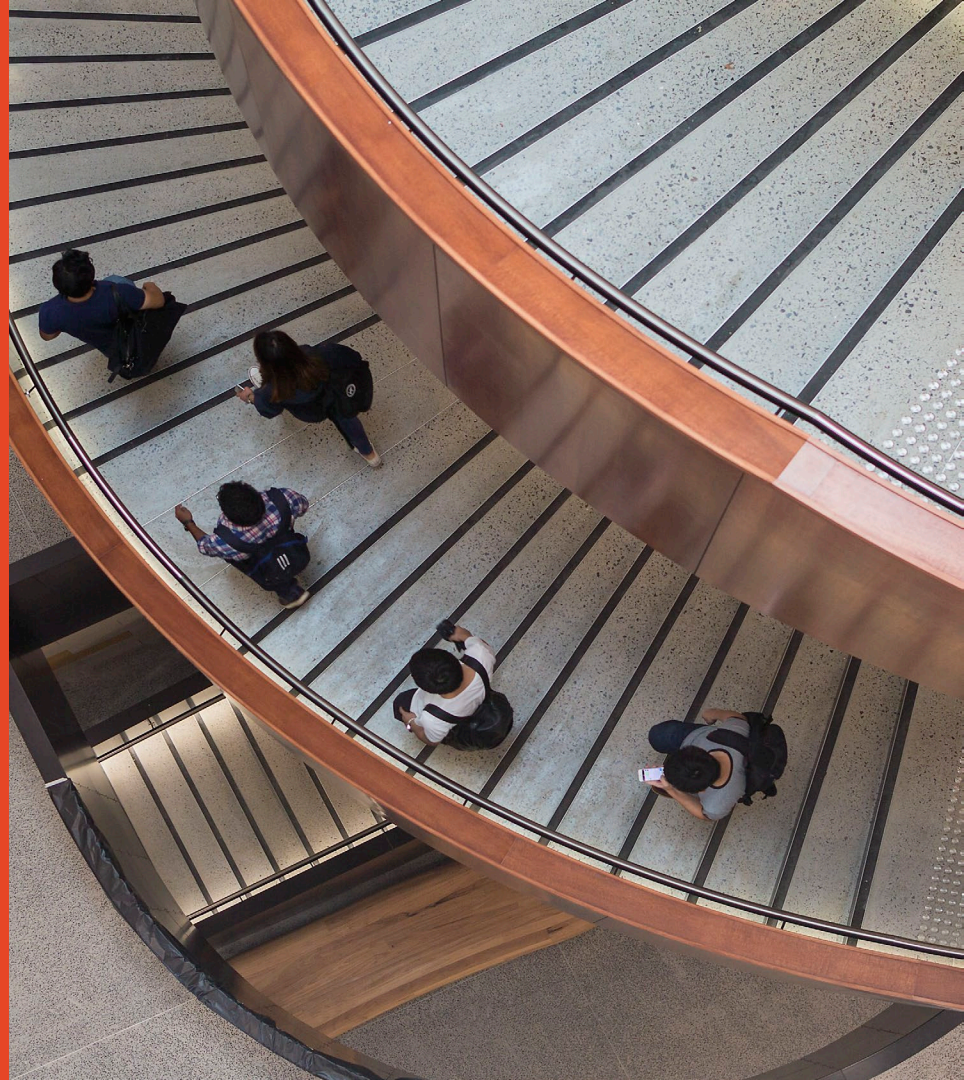

Week 5 Workshop Quiz (Financial Mathematics II) Time – 10 minutes

Module 4 – Debt and Equity Valuation

Brief Recap

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Bonds: An Overview

Bonds – are a loan to a company

Bondholders receive interest payments (coupons) and then their initial investment (principal) at the end of the term.

Important Components are:

1. Maturity Date
2. Face Value (or principal)
3. Coupon Rate
4. Yield to Maturity (or market required rate of return)

Bond Valuation

-Cash flows for a bond are:

$$D = C\left[\frac{1-\left(1+r_d\right)^{-n}}{r_d}\right] + \frac{F}{\left(1+r_d\right)^n}$$

C = Coupon payment

F = Face value

n = Number of periods until maturity

rd = Required rate of return on debt

Bond relationships

1. The value of a bond is inversely related to changes in the market's required yield to maturity.
2. The market value of a bond will be: Less than the face value (discount bond), if the market's required yield to maturity is above the coupon interest rate. Greater than face value (premium bond), if the market's required yield to maturity is below the coupon interest rate.
3. As the maturity date approaches, the market value of a bond approaches its face value.
4. Long-term bonds have greater interest rate risk than short-term bonds.

Stocks: An Overview

-**Stocks –** ownership position in the corporation.

Make returns from stocks through:

1. Dividends
2. Capital Gains

Here we focus on dividends as future payments of the stock

Dividend Discount Model

-Cash flows for a stock are:

$$PV = \frac{D}{r}$$

D = Dividend payments (constant)

r = Required rate of return

Key idea: Prices should equal future discounted dividends of a stock

Gordon Growth Model

-Companies tend to grow over time

-Cash flows for a growing stock are:

$$P_0 = \frac{d_1}{r_e - g}$$

D_1 = Dividend payments in period 1

r = Required rate of return

g = Growth rate

2-Stage Growth Model

-Companies tend to grow over time in different phases

-Cash flows for a 2-stage growing stock are:

$$PV = \frac{D_0 * (1 + g_1)}{1 + r} + \frac{D_1 * (1 + g_1)}{(1 + r)^2} + \frac{D_2 * (1 + g_2)}{(1 + r)^2} + \cdots$$

D_1 = Dividend payments in period 0

r = Required rate of return

g_1 = Growth rate in Stage 1

g_2 = Growth rate in Stage 2

2-Stage Growth Model (cont)

Solve in three-steps:

1. Calculate the dividends for growth stage 1

$$D_1 = D_0 * (1 + g_1)$$

$$D_2 = D_1 * (1 + g_1)$$

2. Use GGM to calculate the PV for growth stage 2

$$PV_{g_2} = \frac{D_{2*}(1 + g_2)}{r - g_2}$$

3. Discount Growth Stage 2 and add the cash flows

$$PV_0 = \frac{D_1}{1 + r} + \frac{D_2}{(1 + r)^2} + \frac{\frac{D_3}{r - g_2}}{(1 + r)^2}$$

PVGO

Stock prices can be decomposed into two components:

1. The present value of earnings under a no-growth policy
2. Present value of growth opportunities (PVGO)

The **present value of growth opportunities (PVGO)** is the amount that investors are willing to pay for growth.

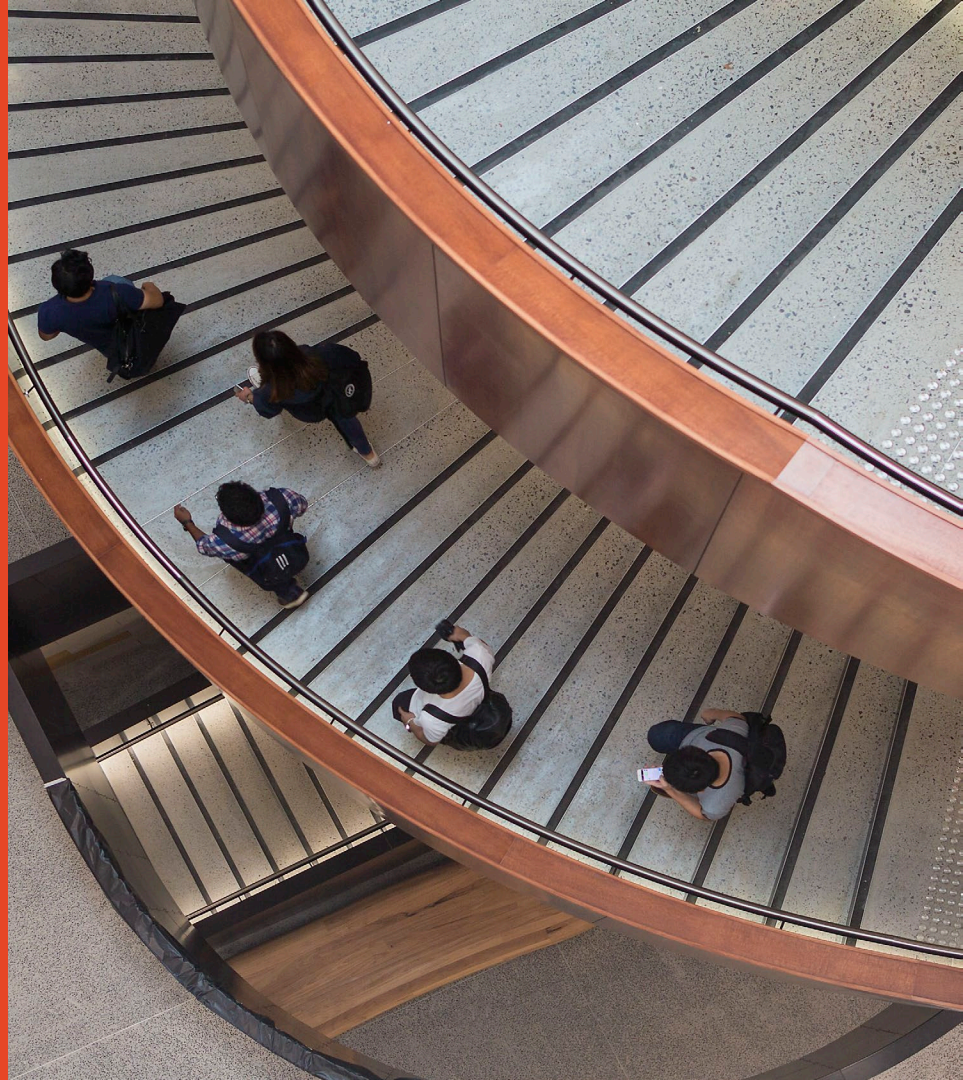
The formula for PVGO is: $P = \frac{EPS}{r} + PVGO$

P = Current stock price

EPS_1 = EPS for next year

$PVGO$ = Present value of growth opportunities

Workshop Questions



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Question One

What would be the effect on the price of bonds of (1) an unexpected rise in interest rates, and (2) an unexpected fall in interest rates? Would these interest rate changes have a similar effect on the price of shares? Explain.

Question Two

Suppose the current Commonwealth Bond yield is 5.5% p.a. compounded semi-annually. Which of the following risk-free bonds would have the lowest price?

a) *A zero-coupon bond, $100 face value, 12 years until maturity*

b) *A 6.25% p.a. semi-annual coupon bond, $100 face value, 3 years to maturity*

c) *A 5.5% p.a. semi-annual coupon bond, $100 face value, 5 years to maturity*

d) *A 10% p.a. annual coupon bond, $100 face value, 8 years to maturity*

e) *A 8.5% p.a. quarterly coupon bond, $100 face value, 10 years to maturity*

Question Three

Devonport Group Ltd is planning on raising new debt capital through the issue of 9% p.a. semi-annual coupon bonds with a face value of $100. These bonds will mature in five years. If the current market yield is 11% p.a. compounded semi-annually, how much will each bond cost?

Question Four

Ventracorp Ltd is planning on raising $20m in new debt through a new corporate bond issuance. The bonds have a coupon rate of 12%pa with coupons paid on a quarterly basis, a face value of $100 and 5 years to maturity. Similar bonds are priced with a yield of 10.125% pa compounded semi-annually. If Ventracorp is planning on issuing the bonds at a discount of 7% off the fair value, how many bonds must they issue?

Question Five

You wish to value a company that paid a dividend of $1.23 per share yesterday. Furthermore, you are aware that five years ago the company paid a dividend of $0.80 and in the interim years the dividend paid has grown at a constant rate.

a) *What has the annual growth rate in dividends been during the past five years?*

b) *If dividends are expected to grow at the same annual growth rate for the foreseeable future, what is the maximum price that you would be willing to pay for a share in the company if you require a return on equity of 15% p.a.?*

Question Six

ABC has recently announced that it has made a major breakthrough in the development of a new mine. ABC has not paid any dividends in the past. However, with this breakthrough the company has announced that it expects to be able to pay its first dividend of $0.8 in four years' time from now and thereafter the dividend is expected to grow at 5% pa in perpetuity. ABC shareholders require a 12% pa return on their investment. What should be the current price of ABC shares?

Question Seven

DEF pays dividends to its shareholders twice a year, in August and February each year. The August dividend of $0.6 was paid today. The next dividend is due in February next year. The dividend is expected to grow by 3% pa compounded semi-annually forever. Shareholders in DEF require a return of 15% pa compounded annually. How much should one share of DEF be worth today?

Question Eight

Intelligent Student (INT) Ltd has forecast the following EPS for the next 5 years:

Year 1	Year 2	Year 3	Year 4	Year 5
-$1.20	-$0.45	$0.50	$1.40	$1.90

The company then expects EPS to grow at 5% pa each year in perpetuity after year 5. The company maintains a constant payout ratio of 50% of earnings each year. The company pays only one dividend per year. Shareholders require a return of 12% pa compounded semi-annually. What should be the price of one Intelligent Student (INT) Ltd share today?

Question Nine

Ventracorp Ltd recently paid a dividend of $0.20 per share. The dividend is expected to grow at 25% for the next 4 years and then revert back to a long-term growth rate of 3% pa forever after that. If the company's shareholders require a return of 12% p.

a) How much should one share of Ventracorp cost today?

b) How much should one share in Ventracorp cost exactly 3 years?

Question One

What would be the effect on the price of bonds of (1) an unexpected rise in interest rates, and (2) an unexpected fall in interest rates? Would these interest rate changes have a similar effect on the price of shares? Explain.

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What would be the effect on the price of bonds of (1) an unexpected rise in interest rates, and (2) an unexpected fall in interest rates? Would these interest rate changes have a similar effect on the price of shares? Explain.

(1) An unexpected rise in interest rates would have a negative effect on the price of bonds (i.e. it will reduce the market price) due to the inverse relationship between interest rates and bond prices.

This is best demonstrated by way of example. Consider the impact of an increase in interest rates to 10% p.a. on a bond with a coupon rate of 8% p.a. If a new bond of similar risk profile were issued with coupon set at the new market interest rate of 10%, then investors would not be willing to pay the same price for the first bond, as its future cash flows (based on coupon multiplied by face value) will be lower. These future cash flows will be discounted at the new market interest rate, so they will have a lower present value, which translates to a lower market price of the bond.

Question One

What would be the effect on the price of bonds of (1) an unexpected rise in interest rates, and (2) an unexpected fall in interest rates? Would these interest rate changes have a similar effect on the price of shares? Explain.

(20) An unexpected fall in interest rates will have the opposite effect for analogous reasons.

Changes in interest rates would have a similar effect on the price shares. This is because the value of share is the present value of all future dividends. These dividends are discounted to find the present value. This discount rate reflects the opportunity cost of capital. If interest generally were to increase, this increases the opportunity cost and increases the rate of return required by shareholders – if they don't get that rate of return, they will switch their investments to debt. Since future cash flows are being discounted more heavily, the price of shares is expected to fall when interest rates increase, and vice versa.

Question Two

Suppose the current Commonwealth Bond yield is 5.5% p.a. compounded semi-annually. Which of the following risk-free bonds would have the lowest price?

a) *A zero-coupon bond, $100 face value, 12 years until maturity*

b) *A 6.25% p.a. semi-annual coupon bond, $100 face value, 3 years to maturity*

c) *A 5.5% p.a. semi-annual coupon bond, $100 face value, 5 years to maturity*

d) *A 10% p.a. annual coupon bond, $100 face value, 8 years to maturity*

e) *A 8.5% p.a. quarterly coupon bond, $100 face value, 10 years to maturity*

Question Two

This question does not require any calculations. We know by the relationship between bond prices and yields that a) is at a **discount** (since there is only a face value), b) is priced at a **premium** (since the coupon rate is higher than the yield, c) is priced at its **par value** (since the coupon rate equals the yield), d) is priced at a **premium** (since the coupon rate is higher than the yield) and e) is priced at a **premium** (since the coupon rate is higher than the yield). Therefore, the one with the lowest price is a)

Question Three

Devonport Group Ltd is planning on raising new debt capital through the issue of 9% p.a. semi-annual coupon bonds with a face value of $100. These bonds will mature in five years. If the current market yield is 11% p.a. compounded semi-annually, how much will each bond cost?

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Devonport Group Ltd is planning on raising new debt capital through the issue of 9% p.a. semi-annual coupon bonds with a face value of $100. These bonds will mature in five years. If the current market yield is 11% p.a. compounded semi-annually, how much will each bond cost?

In this instance the coupon payments and the yield are both compounded semi-annually and hence we can just use the present value of a bond equation:

$$Bond\ price = \$4.50\left[\frac{1-(1.055)^{-10}}{0.055}\right] + \frac{100}{(1.055)^{10}} = \$92.46$$

Question Four

Ventracorp Ltd is planning on raising $20m in new debt through a new corporate bond issuance. The bonds have a coupon rate of 12%pa with coupons paid on a quarterly basis, a face value of $100 and 5 years to maturity. Similar bonds are priced with a yield of 10.125% pa compounded semi-annually. If Ventracorp is planning on issuing the bonds at a discount of 7% off the fair value, how many bonds must they issue?

Question Four

Ventracorp Ltd is planning on raising $20m in new debt through a new corporate bond issuance. The bonds have a coupon rate of 12%pa with coupons paid on a quarterly basis, a face value of $100 and 5 years to maturity. Similar bonds are priced with a yield of 10.125% pa compounded semi-annually. If Ventracorp is planning on issuing the bonds at a discount of 7% off the fair value, how many bonds must they issue?

In this case the coupon payments and the yield on the bond do not match. So we must first calculate the effective yield per quarter to match the quarterly coupon payments:

$$ER = \left(1 + \frac{0.10125}{2}\right)^{2x0.25} - 1 = 0.025$$

Then use the bond formula to determine the price of a single bond:

$$Bond\ price = \$3\left[\frac{1-(1.025)^{-20}}{0.025}\right] + \frac{100}{(1.025)^{20}} = \$107.795$$

Question Four

Ventracorp Ltd is planning on raising $20m in new debt through a new corporate bond issuance. The bonds have a coupon rate of 12%pa with coupons paid on a quarterly basis, a face value of $100 and 5 years to maturity. Similar bonds are priced with a yield of 10.125% pa compounded semi-annually. If Ventracorp is planning on issuing the bonds at a discount of 7% off the fair value, how many bonds must they issue?

Then calculate the discounted price of the bond:

Discounted bond price = $107.795 x 0.93 = $100.249

Finally determine the number of bonds required:

Number of bonds required = $20,000,000/$100.249 = 199,504 bonds

Question Five

You wish to value a company that paid a dividend of $1.23 per share yesterday. Furthermore, you are aware that five years ago the company paid a dividend of $0.80 and in the interim years the dividend paid has grown at a constant rate.

a) *What has the annual growth rate in dividends been during the past five years?*

b) *If dividends are expected to grow at the same annual growth rate for the foreseeable future, what is the maximum price that you would be willing to pay for a share in the company if you require a return on equity of 15% p.a.?*

Question Five

a) We can use the FV of a single cash flow formula to find the relationship between the future value and present value based on a growth rate and a number of periods.

$$FV = PV\left(1+g\right)^n \therefore g = \sqrt[n]{\frac{FV}{PV}} - 1 = \sqrt[5]{\frac{1.23}{0.80}} - 1 = 9\%$$

b) The current dividend (d_0) is \$1.23. We need to use the next dividend (d_1) in calculating the constant-dividend growth model.

$$P_t = \frac{d_1}{r_e - g} = \frac{1.23\left(1.09\right)}{0.15 - 0.09} = \frac{1.34}{0.15 - 0.09} = \$22.33$$

Question Six

ABC has recently announced that it has made a major breakthrough in the development of a new mine. ABC has not paid any dividends in the past. However, with this breakthrough the company has announced that it expects to be able to pay its first dividend of $0.8 in four years' time from now and thereafter the dividend is expected to grow at 5% pa in perpetuity. ABC shareholders require a 12% pa return on their investment. What should be the current price of ABC shares?

Question Six

This can be calculated as a deferred perpetuity. From year four onwards you have constant growth so using the constant dividend growth model you can determine the price in the period PRIOR to the first dividend (year 3):

$$P_3 = \frac{0.8}{0.12 - 0.05} = \$11.43$$

Then you discount that price back to time zero using the present value of a single cash flow equation:

$$P_0 = \frac{11.43}{(1 + 0.12)^3} = \$8.14$$

Question Seven

DEF pays dividends to its shareholders twice a year, in August and February each year. The August dividend of $0.6 was paid today. The next dividend is due in February next year. The dividend is expected to grow by 3% pa compounded semi-annually forever. Shareholders in DEF require a return of 15% pa compounded annually. How much should one share of DEF be worth today?

Question Seven

Since the dividends are paid semi-annually we must first determine the effective half-yearly required return on equity (using the effective rate equation):

$$ER = (1 + 0.15)^{1/2} - 1 = 0.072381$$

Then we can calculate the share price using the constant dividend growth formula (noting the dividend given is D_0). The effective growth rate on the dividend is simply 3% divided by 2 = 1.5% per six months.

$$P_0 = \frac{0.6\ (1+0.015)}{0.072381-0.015} = \$10.61$$

Question Eight

Intelligent Student (INT) Ltd has forecast the following EPS for the next 5 years:

Year 1	Year 2	Year 3	Year 4	Year 5
-$1.20	-$0.45	$0.50	$1.40	$1.90

The company then expects EPS to grow at 5% pa each year in perpetuity after year 5. The company maintains a constant payout ratio of 50% of earnings each year. The company pays only one dividend per year. Shareholders require a return of 12% pa compounded semi-annually. What should be the price of one Intelligent Student (INT) Ltd share today?

Question Eight

To answer this question, we first need to determine the DPS each year. This is done by taking the DPS share and multiplying it by the 50% payout ratio to determine the DPS, HOWEVER, the dividends CANNOT be negative, so any negative result would mean a dividend of zero. This gives the dividends per share over the first five years in the table below:

Year 1	Year 2	Year 3	Year 4	Year 5
$0	$0	$0.25	$0.70	$0.95

We then calculate the shareholders effective required return per year using the effective rate equation:

$ER = (1+0.12_{/2})^2 - 1 = 0.1236$

Question Eight

We then use the constant dividend growth model to find the share price after we have constant growth (after year 5)

$$P_5 = \frac{0.95\,(1 + 0.05)}{0.1236 - 0.05} = \$13.55299$$

Finally, we discount the future dividends and the share price in year 5 back to time zero:

$$P_0 = \frac{0.25}{(1+0.1236)^3} + \frac{0.70}{(1+0.1236)^4} + \frac{0.95+13.55299}{(1+0.1236)^5} = \$8.71$$

Question Nine

Ventracorp Ltd recently paid a dividend of $0.20 per share. The dividend is expected to grow at 25% for the next 4 years and then revert back to a long-term growth rate of 3% pa forever after that. If the company's shareholders require a return of 12% p.

a) How much should one share of Ventracorp cost today?

b) How much should one share in Ventracorp cost exactly 3 years?

Question Nine

Part a) Firstly, determine the dividends in the period of the first growth rate (25%). These are given below:

Div1 = 0.2 x 1.25 = 0.25

Div2 = 0.2 x 1.25^2 = 0.3125

Div3 = 0.2 x 1.25^3 = 0.391

Div4 = 0.2 x 1.25^4 = 0.488

Div5 = 0.2 x 1.25^4 x 1.03 = 0.503

We then use the constant dividend growth model to find the share price after we have constant growth (after year 4)

$$P_4 = \frac{0.503}{0.12-0.03} = \$5.589$$

Question Nine

Finally, we discount the future dividends and the share price in year 4 back to time zero:

$$P_0 = \frac{0.25}{(1.12)^1} + \frac{0.3125}{(1.12)^2} + \frac{0.391}{(1.12)^3} + \frac{0.488+5.589}{(1.12)^4} = \$4.61$$

Question Nine

Part b) The share price in year 3 will be the present value of the FUTURE cash flows from that point onwards. So, in this case the future cash flows are the dividend in year 4 and the share price in year 4. These would then need to be discounted back one year to year 3:

$$P_3 = \frac{0.488+5.589}{(1.12)^1} = \$5.43$$

The End

Next week we begin to examine Risk-adjusted Discounted Cash Flows, Investor Preferences, Diversification and Portfolios